OBJECT ORIENTED SOFTWARE ENGINEERING

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Bottom-up Integration Testing

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Bottom Up Integration:

- Testing takes place from bottom of the control flow, **step by step to upwards**.

- Here the higher level modules **might not be developed by that time lower level modules are tested**. Hence to overcome this issue here we are using drivers which are **dummy calling programs**.

Bottom up Integration…

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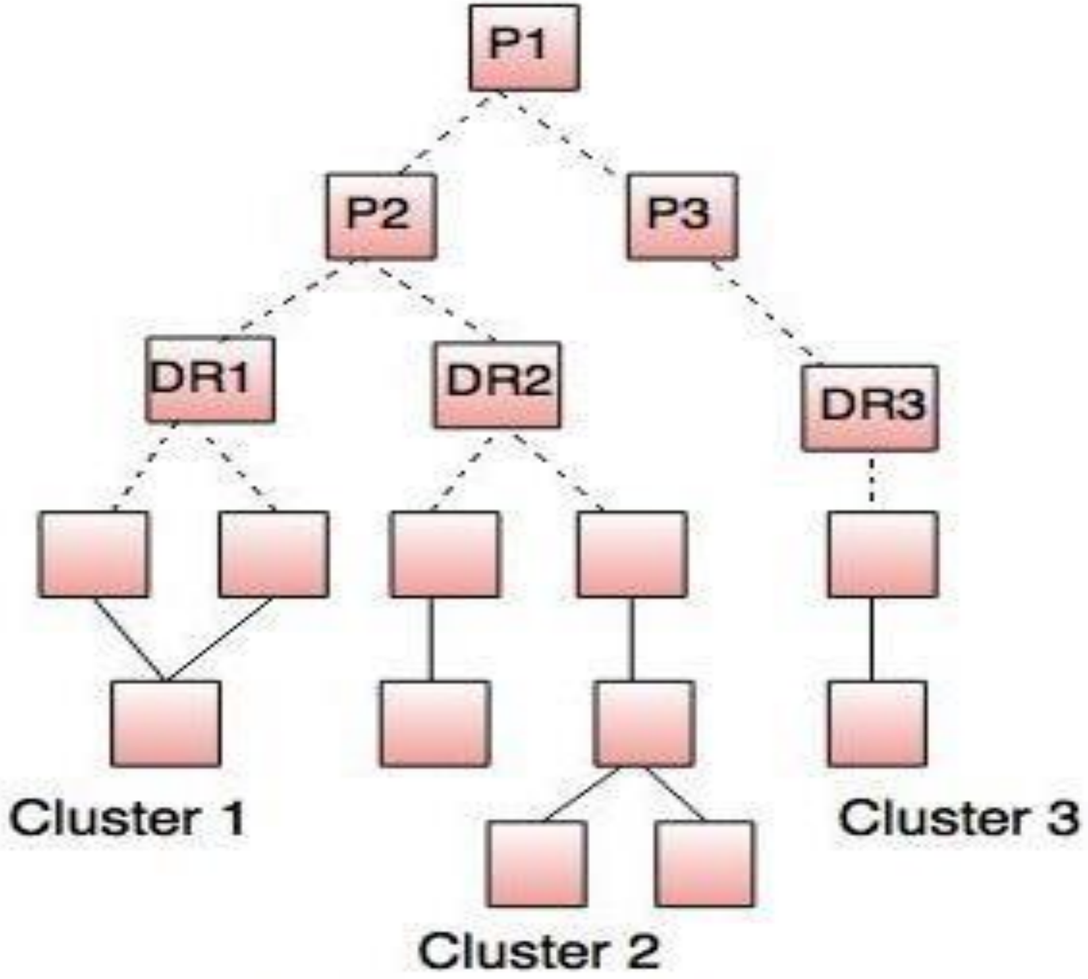

Fig. - Bottom-up integration

Bottom up Integration Testing…

- In bottom up integration testing the components are combined from the lowest level in the program structure.

The bottom-up integration is implemented in following steps:

1. The low level components are merged into **clusters** which perform a specific software sub function.
2. A control program for testing(**driver**) coordinate test case input and output.
3. After these steps are **tested in cluster**.
4. The **driver is removed** and **clusters are merged by moving upward** on the program structure.

Bottom Up Integration Testing

- **Advantages:**
- Efficient application
- Less time requirements
- Test conditions are easier to create

- **Disadvantages**
- Requires several drivers
- Data flow is tested late
- Poor support for early release
- Key interface defects are detected

THANK YOU

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